INJECTIVE AND SURJECTIVE MAPS OF PRESHEAVES ARE THE MONOMORPHISMS AND EPIMORPHISMS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Injective and surjective maps of presheaves. Same chapter, Definition definition-presheaves-injective-surjective, recorded as the context of this card. Supplies the two notions compared with monomorphisms and epimorphisms.
- Representable presheaves and the Yoneda identification. Same chapter, the identification of the elements of a presheaf at an object with natural transformations from the representable presheaf, used at source lines 389-396. Turns elements of a value into natural transformations so that monicity can be applied.
- Pushouts in the category of sets. Standard: the pushout of a map of sets along itself has two canonical maps from the target, which coincide exactly when the map is surjective. Produces the two natural transformations whose equality forces surjectivity.

Source reference. Original equation 090F: sources/stacks-sites/pinned-original-source.tex, source lines 390–390. Navigation pointer invoked at proof line 390; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Definition 0.1. Let $\mathcal{C}$ be a category, and let $\varphi : \mathcal{F} \to \mathcal{G}$ be a map of presheaves of sets.

(1) We say that φ is *injective* if for every object U of $\mathcal{C}$ the map $\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)$ is injective.
(2) We say that φ is *surjective* if for every object U of $\mathcal{C}$ the map $\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)$ is surjective.

Lemma 0.2. *The injective (resp. surjective) maps defined above are exactly the monomorphisms (resp. epimorphisms) of $PSh(\mathcal{C})$. A map is an isomorphism if and only if it is both injective and surjective.*

Proof. We shall show that $\varphi : \mathcal{F} \to \mathcal{G}$ is injective if and only if it is a monomorphism of $PSh(\mathcal{C})$. Indeed, the "only if" direction is straightforward, so let us show the "if" direction. Assume that φ is a monomorphism. Let $U \in \mathrm{Ob}(\mathcal{C})$; we need to show that φ_U is injective. So let $a, b \in \mathcal{F}(U)$ be such that $\varphi_U(a) = \varphi_U(b)$; we need to check that $a = b$. Under the isomorphism (090F), the elements a and b of $\mathcal{F}(U)$ correspond to two natural transformations $a', b' \in \mathrm{Mor}_{PSh(\mathcal{C})}(h_U, \mathcal{F})$. Similarly, under the analogous isomorphism $\mathrm{Mor}_{PSh(\mathcal{C})}(h_U, \mathcal{G}) = \mathcal{G}(U)$, the two equal elements $\varphi_U(a)$ and $\varphi_U(b)$ of $\mathcal{G}(U)$ correspond to the two natural transformations $\varphi \circ a', \varphi \circ b' \in \mathrm{Mor}_{PSh(\mathcal{C})}(h_U, \mathcal{G})$, which therefore must also be equal. So $\varphi \circ a' = \varphi \circ b'$, and thus $a' = b'$ (since φ is monic), whence $a = b$. This finishes (1).

We shall show that $\varphi : \mathcal{F} \to \mathcal{G}$ is surjective if and only if it is an epimorphism of $PSh(\mathcal{C})$. Indeed, the "only if" direction is straightforward, so let us show the "if" direction. Assume that φ is an epimorphism.

For any two morphisms $f : A \to B$ and $g : A \to C$ in the category *Sets*, we let $\mathrm{inl}_{f,g}$ and $\mathrm{inr}_{f,g}$ denote the two canonical maps from B and C to $B \coprod_A C$. (Here, the pushout is evaluated in *Sets*.)

Now, we define a presheaf $\mathcal{H}$ of sets on $\mathcal{C}$ by setting $\mathcal{H}(U) = \mathcal{G}(U) \coprod_{\mathcal{F}(U)} \mathcal{G}(U)$ (where the pushout is evaluated in *Sets* and induced by the map $\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)$) for every $U \in \mathrm{Ob}(\mathcal{C})$; its action on morphisms is defined in the obvious way (by the functoriality of pushout). Then, there are two natural transformations $i_1 : \mathcal{G} \to \mathcal{H}$ and $i_2 : \mathcal{G} \to \mathcal{H}$ whose components at an object $U \in \mathrm{Ob}(\mathcal{C})$ are given by the maps $\mathrm{inl}_{\varphi_U, \varphi_U}$ and $\mathrm{inr}_{\varphi_U, \varphi_U}$, respectively. The definition of a pushout shows that $i_1 \circ \varphi = i_2 \circ \varphi$, whence $i_1 = i_2$ (since φ is an epimorphism). Thus, for every $U \in \mathrm{Ob}(\mathcal{C})$, we have $\mathrm{inl}_{\varphi_U, \varphi_U} = \mathrm{inr}_{\varphi_U, \varphi_U}$. Thus, φ_U must be surjective (since a simple combinatorial argument shows that if $f : A \to B$ is a morphism in *Sets*, then $\mathrm{inl}_{f,f} = \mathrm{inr}_{f,f}$ if and only if f is surjective). In other words, φ is surjective, and (2) is proven.

We shall show that $\varphi : \mathcal{F} \to \mathcal{G}$ is both injective and surjective if and only if it is an isomorphism of $PSh(\mathcal{C})$. This time, the "if" direction is straightforward. To prove the "only if" direction, it suffices to observe that if φ is both injective and surjective, then φ_U is an invertible map for every $U \in \mathrm{Ob}(\mathcal{C})$, and the inverses of these maps for all U can be combined to a natural transformation $\mathcal{G} \to \mathcal{F}$ which is an inverse to φ. □